fygenson-et-al-2024-the-arrangement-of-marks-impacts-a

Auto-assembled report. Section content is drawn from the summarization pipeline outputs.

Table of Contents

Paper Synthesis

Not available for this paper.

Abstract

Preserved from source without summarization.

Abstract--Data visualizations present a massive number of potential messages to an observer. One might notice that one group's average is larger than another's, or that a difference in values is smaller than a difference between two others, or any of a combinatorial explosion of other possibilities. The message that a viewer tends to notice - the message that a visualization 'affords' - is strongly affected by how values are arranged in a chart, e.g., how the values are colored or positioned. Although understanding the mapping between a chart's arrangement and what viewers tend to notice is critical for creating guidelines and recommendation systems, current empirical work is insufficient to lay out clear rules. We present a set of empirical evaluations of how different messages-including ranking, grouping, and part-to-whole relationships-are afforded by variations in ordering, partitioning, spacing, and coloring of values, within the ubiquitous case study of bar graphs. In doing so, we introduce a quantitative method that is easily scalable, reviewable, and replicable, laying groundwork for further investigation of the effects of arrangement on message affordances across other visualizations and tasks. Pre-registration and all supplemental materials are available at https://osf.io/np3q7 and https://osf.io/bvy95 respectively.

Introduction (Paper-Level)

Not available for this paper.

General Discussion

Not available for this paper.

Critique

Not available for this paper.